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May 26, 1967
INTERIM REVISION OF
Fed. Spec. DDD-T-501c
August 27, 1953 and
Fed. Spec. CCC-T-561
June 28, 1932

# INTERIM FEDERAL SPECIFICATION

## TOWELS AND TOWELING, COTTON FOR GLASSWARE

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D. C., 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

The General Services Administration has authorized Federal agencies to use this Interim Federal Specification as a valid exception to Federal Specification DDD-T-501c, dated August 27, 1953 and CCC-T-561, dated June 28, 1932.

### 1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers bleached cotton towels and bleached cotton toweling, primarily intended to be used for drying glassware.

1.2 Classification.

1.2.1 Types. Towels and toweling covered by this specification shall be of one type and the following styles as specified, (see 6.2).

Style a. Towel, Cotton
Style b. Toweling, Cotton.

### 2. APPLICABLE DOCUMENTS.

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

Federal Specifications:

V-T-276 - Thread, Cotton.
CCC-T-191 - Textile Test Methods.

Federal Standards:

Fed. Std. No. 4 - Glossary of Fabric Imperfections.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 751 - Stitches, Seams, and Stitching.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies).

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Indentification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D. C.).

Standard Color Card of America

(Copies may be obtained from the Color Association of the United States, Inc., 200 Madison Avenue, New York, New York).

3. REQUIREMENTS.

3.1 Material.

3.1.1 Cotton. The warp and filling yarns shall be of thoroughly cleaned cotton, free from waste.

3.1.2 Thread. The thread used for hemming style a towels shall be ticket number 40 or 50, 3-ply, for the needle and 60, 3-ply, for the bobbin and shall conform to type IA2 of V-T-276.

3.2 Physical requirements. The style a towels and style b cloth shall conform to the requirements shown in table I when tested as specified in 4.3.

Table I. Physical requirements

| Styles | Weight per square yard (minimum) | Yarns per inch (minimum) | | Breaking Strength (min.) | |
|---|---|---|---|---|---|
| | | Warp | Filling | Warp | Filling |
| | Ounces | | | pounds | pounds |
| a and b | 5.1 | 54 | 38 | 50 | 50 |

3.2.1 Weave. Style a towels and style b cloth, shall be of a plain weave.

3.2.2 Finish. The cloth for the style a towels and the style b cloth shall be bleached white.

3.3 Design.

3.3.1 Styles a and b. Both styles shall have either a checked pattern formed by two colored ends and two colored picks woven at uniform intervals of 1/2 to 1 inch apart; or a warp stripe pattern consisting of stripes containing two colored ends woven parallel to the length at uniform intervals of 1/2 to 1 inch.

3.4 Color. The cloth for style a towels and style b cloth, shall be bleached white as specified in 3.2.2; except for the colored yarns which shall be an approximate match of Cable No. 70180 (Old Glory Red) or Cable No. 70087 (Royal Blue) of the Standard Color Card of America.

3.4.1 Colorfastness. The colored yarns in the finished towels and towel cloth shall show good colorfastness to laundering and chlorine bleaching and shall not stain the white part of the towels when tested as specified in 4.3.



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3.5 Hems and selvages. The finished style a towels shall have fast selvages on each side and a 1/4 to 3/8 inch hem on each end, with no raw edges showing. Hemming shall be done with 9 to 12 stitches per inch, using stitch type 301, and stitching type EFb-1 of Fed. Std. No. 751. At the beginning and end of each hem, stitching shall be backstitched at least 1/2 inch.

3.6 Size. The finished measurements of the towels and towel cloth shall be as follows:

| Style | Length inches (minimum) | Width, inches 1/ |
|---|---|---|
| a | 30 | 16 |
| b | See 3.7 | 16 |

1/ The width shall be inclusive of selvages, and the following tolerance shall be permitted: plus 1 inch and a minus of 1/2 inch.

3.7 Style b put-up. Unless otherwise specified (6.2), bolts of fifty yards, plus or minus two and one-half yards, of style b cloth shall be put up at the supplier's option by winding into rolls on a flat chipboard or cardboard insert of not less than 1/8 inch in thickness by 12 inches wide, or by folding open width in one yard flat folds.

3.8 Marking for identification. When identification marking is required, the marking requirement shall be as specified (6.2). The marking, when required, shall be clearly legible, shall show good colorfastness to laundering and chlorine bleaching, and shall show no evidence of bleeding into the surrounding cloth.

3.9 Fiber identification. Style a towels and each roll or bolt of style b cloth shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

3.10 Workmanship. The finished towels and cloth shall conform to the quality established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels specified herein.

4. QUALITY ASSURANCE PROVISIONS.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and servides conform to prescribed requirements.

4.2 Inspection. Inspection shall be in accordance with MIL-STD-105, except where otherwise indicated herein.

4.2.1 Inspection of components. In accordance with 4.1 above, the supplier is responsible for insuring that components and materials are manufactured, tested and inspected to ensure conformance with the requirements of this specification.

4.2.2 Examination of the style a towel. The towels shall be examined on both sides. The occurrence of the same defect shall be tallied for one side only. The defects found during the examination shall be classified in accordance with the defects listed in table II. The inspection level shall be I and the acceptable quality level (AQL) shall be 2.5 major defects and 6.5 total defects (major and minor combined) per 100 units. The sample unit shall be one towel. The lot size shall be expressed in units of one towel each.



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Table II. Classification of defects

| Examine | Defect | Classification Major | Minor |
|---|---|---|---|
| Quality of material | Cut, hole or tear. | X | |
| | Smash. | X | |
| | Float and skip: | | |
| | Multiple 3/4 inch or more in combined directions. | X | |
| | Multiple more than 1/4 inch but less than 3/4 inch in combined directions. | | X |
| | Single 4 inches or more in length. | | X |
| | Broken or missing yarn: | | |
| | Two or more contiguous. | X | |
| | Single 4 inches or more in length. | | X |
| | Fine or light filling bar or light place. | X | |
| | Hitchback: | | |
| | Opening between yarns 1/16 inch or more at widest place. | | X |
| | Slub, slug or gout, jerked-in filling or slough off more than 1/8 inch in thickness. | | X |
| | Spot, stain or streak over 1/8 inch in longest dimension clearly visible when viewed at a normal inspection distance (approximately 3 feet). | | X |
| | Selvage: | | |
| | Cut, torn, broken or omitted. | X | |
| | Curled, scalloped or folded. | | X |
| | Not bleached. | | X |
| | Incorrect yarn colors. | | X |
| | Interval of colored yarns less than 1/2 inch or more than 1 inch. | | X |
| | Objectionable odor. | | X |
| Workmanship | Hem less than 1/4 inch or more than 3/8 inch wide. | | X |
| | Hem omitted. | X | |
| | Raw edge showing. | | X |
| | Less than 9 or more than 12 stitches per inch. | | X |



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Table II. Classification of defects (cont'd)

| Examine | Defect | Classification Major | Classification Minor |
|---|---|---|---|
| Workmanship (cont'd) | Wrong stitch or stitching type | | X |
| | Open stitching over 1 inch in length. | X | |
| | Open stitching up to and including 1 inch in length. | | X |
| | Loose stitch tension resulting in loose stitching. | | X |
| | Tight stitch tension (stitches break when normal strain is applied to stitching). | | X |
| | Backstitching omitted (required at the beginning and end of hem). | | X |
| | Backstitched less than 1/2 inch. | | X |
| | Needle chew. | | X |
| Marking(when specified) | Omitted, incomplete, illegible, incorrect or misplaced. | | X |

4.2.2.1 Dimensional examination. Any finished length or width, inclusive of selvages, measurement taken on the towels which is less than specified shall be classified as a defect. Sample unit and lot size shall be as specified in 4.2.2. The inspection level shall be S-3 and the acceptable quality level shall be 4.0 percent defective.

4.2.3 Examination of Style b cloth. Defects found during this examination shall be classified in accordance with 4.2.3.1 through 4.2.3.3.

4.2.3.1 Yard-by-yard examination. Thirty yards from each piece or roll in the sample shall be examined on one side. The sample shall consist of 25 pieces or rolls taken from 25 containers. All defects as defined in Section I of Federal Standard No. 4 which are clearly noticeable at normal inspection distance (3 feet) shall be scored and assigned demerit points as listed in 4.2.3.1.1, except that only those knots and slubs which exceed the limits shown on figure 1 shall be scored. No linear yard shall be penalized more than 4 points. The lot shall be unacceptable if the points per 100 square yards exceed 40.0 points. Point computation shall shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3600}{\text{Contracted width of cloth (inches)} \times 750} = \text{Points per 100 square yards}$$

4.2.3.1.1 Demerit points. Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension - one point

For defects exceeding 3 inches but not exceeding 6 inches in any dimension - two points

For defects exceeding 6 inches but not exceeding 9 inches in any dimension - three points

For defects exceeding 9 inches in any dimension - four points



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Note: The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth
Width less than specified
Not bleached
Color of yarns, when applicable, not as specified
Interval of colored yarns, when applicable, less than 1/2 inch or more than 1 inch.

4.2.3.2 Examination for length.

4.2.3.2.1 Individual roll (or piece). During the yard-by-yard examination, each roll (or piece) in the sample shall be examined for length. Any length found to be less than the minimum specified, more than the maximum specified (when applicable) or more than 2 yards less than the length marked on the ticket, shall be considered a defect with respect to length. The lot shall be unacceptable if two or more rolls (or pieces) in the sample are defective in respect to length.

4.2.3.2.2 Total yardage in sample. The lot shall be unacceptable if the total of the actual lengths of rolls (or pieces) in the sample is less than the total of the lengths marked on the tickets.

4.3 Testing of the end item. The methods of testing specified in CCC-T-191, wherever applicable and as listed in table III, shall be followed. The physical and chemical values specified in section 3 apply to the average of determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit for testing shall consist of 3 towels. The sample size (number of sample units) shall be as shown in the table below. The lot size shall be expressed in units of one towel. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified.

| Lot size | Sample size |
|---|---|
| 800 or less | 2 |
| 801 to 22,000 | 3 |
| 22,001 and over | 5 |

Table III. Test methods

| Characteristic | Requirement Paragraph | Test Method |
|---|---|---|
| Identification of cotton | 3.1.1 | 1200 1/ |
| Weight per square yard | 3.2 | 5041 |
| Yarns per inch | 3.2 | 5050 |
| Breaking strength | 3.2 | 5100 |
| Weave | 3.2.1 | Visual 2/ |
| Colorfastness to: | | |
| Laundering | 3.4.1 | 5610 |
| Chlorine bleaching | 3.4.1 | 5600 |

1/ A certificate of compliance is acceptable for this requirement.

2/ One determination shall be made from each sample unit and the results reported as "pass" or "fail."



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5. PREPARATION FOR DELIVERY.

5.1 Packaging. Packaging shall be level A, B or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Style a towel. Each towel shall be folded to approximate dimensions of 8 x 15. Twelve towels shall be bundled with six towels reversed. The bundle shall be packaged in a polyethylene film bag, heat sealed; or wrapped in paper conforming to UU-P-268, grade B with all folds, seams and laps secured with tape conforming to PPP-T-76.

5.1.1.2 Style b toweling. Bolts of style b toweling in widths and lengths as required shall be packaged in accordance with PPP-P-51, class 3.

5.1.2 Level B.

5.1.2.1 Style a towel. Each towel shall be folded to approximate dimensions of 8 x 15. Twelve towels shall be bundled with six towels reversed. The bundle shall be packaged in a polyethylene film bag, closed; or wrapped in paper conforming to UU-P-268, grade B with all folds, seams and laps secured with tape.

5.1.2.2 Style b toweling. Bolts of style b toweling in widths and lengths as required shall be packaged in accordance with PPP-P-51, Class 3.

5.1.3 Level C. The towels or toweling shall be packaged in accordance with the suppliers commercial practice.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Style a towel. Twenty-five unit packages of style a towels shall be packed flat in a close-fitting box conforming to PPP-B-636 V3c. The box shall be closed, waterproofed sealed and strapped in accordance with the appendix to the box specification.

5.2.1.2 Style b toweling. Bolts of style b toweling of like description shall be packed in accordance with PPP-P-51.

5.2.2 Level B.

5.2.2.1 Style a towel. Twenty-five unit packages of style a towels shall be packed flat in a close-fitting box conforming to PPP-B-636, class-domestic. The box shall be closed in accordance with the appendix to the box specification.

5.2.2.2 Style b toweling. Bolts of style b toweling of like description shall be packed in a accordance with PPP-P-51.

5.2.3 Level C. The towels or toweling shall be packed to insure carrier acceptance and safe delivery at destination in containers complying with the rules and regulations applicable to the mode of transportation.

5.3 Marking. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Federal Standard No. 123.

6. NOTES.

6.1 Intended use. The towels and toweling covered by this specification are intended primarily for use in drying glassware.



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6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.
(b) Style required (1.2).
(c) Style b put-up, if other than specified (3.7).
(d) Marking for identification when required (3.8).
(e) Selection of applicable levels of packaging and packing (5.1 and 5.2).



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